

LG1 Device Specification

Lightweight industrial Anchor with WiFi



Overview

The LG1 is an indoor **industrial Anchor with WiFi connectivity** to be used as a Ultra-Wideband Real-Time Location System (RTLS) gateway device managing the communication with the central RTLS server. Its robust design, flexibility and adaptability makes it **ideal for tracking and navigation applications in an industrial environment**.



Figure 1: Front and back views of the LG1 device

Key Features

- Supports both TDoA (Time Difference of Arrival) as well as TWR (Two-Way Ranging) location measurement techniques. Location accuracy sub-0.5m or better.
- Backhaul via WiFi 2.4GHz b/g/n IoT.
- Supports both UWB channel 5 (6.5 GHz) and channel 9 (8 GHz) with UWB RF compliance compatible with FCC, Japan's ARIB and ETSI of the CE.
- With Low Noise Amplifier, the device provides superior UWB range coverage.
- Integrated Motion sensor: 3-axis accelerometer.
- One multi-function button.
- RGB front LEDs for status indication.
- Buzzer for alarm functions.
- Firmware update over WiFi and USB.
- Free software configuration and visualization tools (for Android, iOS, Windows, macOS and Linux).
- An open [online documentation](#) and [community forum](#).

Electrical Parameters

Parameter	Value
Certification	FCC, CE, ARIB (Japan)
VDC power supply	7VDC to 32VDC, 3W max
USB (power and data)	5VDC 500mA max
Operating temperature (without battery) ¹	-40°C - +85°C
UWB supported channels	Channel 5 (6240–6739.2 MHz, CF=6489.6 MHz) Channel 9 (7738–8237.2 MHz, CF=7987.2 MHz, FiRa compatible)
UWB transmit powers	FCC/ARIB/ETSI: -41.3 dBm/MHz max
Bluetooth	2.4 GHz BLE 5.2, TX Power up to +8 dBm
WIFI Frequency Range	2.4 GHz IEEE 802.11 b/g/n
WIFI TX Output Power	19.5 dBm
WIFI Encryption	Open, WPA2-PSK, WPA2-PSK-256, WPA3-SAE
WIFI Country Code	01 (world safe)

Mechanical Parameters

Parameter	Value
Size	W=85 x H=57 x D=17 mm
Weight	50g
Material	ABS
Color	White
IP rating	IP30 (1 screw)

Mounting	Standard 1/4" screwball head camera mount
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Device Overview



Figure 2: Connectors view of the device

Safety Instructions

- Do not expose devices to water or moisture while in operation.
- Do not expose devices to excessive heat from any source. The product is designed for reliable operation at industrial temperature¹.

Warnings

- This product shall be connected only to the USB with a minimum current supply of 0.5 A.
- Any external power supply used with this product shall comply with relevant regulations and standards applicable in the country of intended use.
- This product should be operated in a ventilated environment with a non-condensing environment and should not be covered when being operated.
- There are no user-serviceable parts inside the product, and opening the unit will likely damage the product and invalidate the warranty.
- The cables and connectors of all peripherals used with this product must have adequate insulation to meet relevant safety requirements.

Order Information

- Part number: LR-LG1A1
- Custom code: HS 8517.69.9000
- Packaging: 1pcs in paper box, 90 cm x 60 cm x 20 cm 52g

About Us

At LEAPS, we specialize in delivering cutting-edge UWB-based localization technology to help businesses and people locate new values.

Our unique technology and innovative solutions created in close cooperation with leading UWB chip producer Qorvo, coupled with a comprehensive range

of products, provide unmatched performance and versatility. From tags and anchors to custom-designed systems, our UWB hardware and software is built to meet the specific needs of diverse industries.

Contact Us

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